

IAL Physics Unit 5 (WPH15): Rapid Revision

Last-minute summary · 1 h 45 min · 90 marks · 10 MCQ + structured questions + one 6-mark answer

EXAM RULES THAT COST MARKS

- “Show that”: give 1 more s.f. than printed; if stuck, use the printed value next.
- Missing or wrong unit loses the final mark. Use K, never °K.
- $g = 9.81 \text{ N kg}^{-1}$; $1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$ (1.67 was rejected); do not round mass defects.
- “Assess the claim”: your value, then a numerical comparison, then a conclusion.
- Gradient from a large triangle; read the period over several cycles.
- 6-marker: 6 linked points are needed for 6/6. Write 6-7 numbered cause → effect sentences.

THERMAL PHYSICS

$\Delta E = mc\Delta\theta$	$\Delta\theta$ is a difference (°C = K)
$\Delta E = L\Delta m$	state change, T constant
$pV = NkT$	T in K; N = total mass ÷ molecule mass
$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$	$T \times 2 \rightarrow c_{\text{rms}} \times \sqrt{2}$
$U = \frac{3}{2}NkT = \frac{3}{2}pV$	ideal gas; U ≠ pV

- **Internal energy** = sum of the random KE + PE of the molecules.
- **Flat part of a cooling curve**: KE constant, PE decreases.
- **Mixing**: energy lost by hot = mL (melt ice) + $mc(\theta - 0)$. Keep θ as an unknown.
- **Practical L errors**: heat lost → L too big; heater on longer than recorded → L too small.

NUCLEAR

$\lambda t_{1/2} = \ln 2$	match units (s^{-1} with s)
$A = \lambda N$	$N = \text{mass} \div (A \times 1.66 \times 10^{-27})$
$A = A_0 e^{-\lambda t}$	$t = \ln(A/A_0)/(-\lambda)$
$P = A \times E_{\text{decay}}$	energy per decay in J
$I = I_0 e^{-\mu x}$	$\ln I$ vs x : gradient $-\mu$; $x_{1/2} = \ln 2/\mu$

MASS DEFECT → MEV

$\Delta m \text{ (u)} \rightarrow \times 1.66 \times 10^{-27} \rightarrow \times c^2 \text{ (J)} \rightarrow \div 1.60 \times 10^{-13} \text{ (MeV)} \rightarrow \div A \text{ for B.E./nucleon.}$

- **β⁻ equation**: always include $\bar{\nu}_e$. Balance top and bottom numbers.
- **α gets almost all the KE**: momentum is conserved and the daughter is heavier, so it is slower and has less KE. $E_\alpha = (m_{\text{daughter}}/m_{\text{parent}})Q$.
- **β spectrum is continuous**: energy is shared randomly among 3 particles (β, $\bar{\nu}$, nucleus).
- **Random**: cannot predict which nucleus or when. **Spontaneous**: unaffected by external conditions.
- **Cloud chamber**: thickness ↔ ionising ability; straightness ↔ mass.
- **Background**: measure it and subtract it before applying the inverse square law; repeat for random error.
- **B.E./nucleon peaks at Fe-56**; fission and fusion both increase it, so mass decreases and energy is released.

GRAVITATIONAL FIELDS

$F = GMm/r^2$; $g = GM/r^2$	$r = R + h$
$V = -GM/r$	$M = -rV/G$ from a graph
$\Delta E = GMm(1/r_1 - 1/r_2)$	never mgΔh
$T^2 = 4\pi^2 r^3/GM$	from $GMm/r^2 = m\omega^2 r$
$v_{\text{esc}} = \sqrt{2GM/R}$	black hole: $R = 2GM/c^2$

- **Geostationary orbit**: $T = 24 \text{ h}$, so $r = 4.23 \times 10^7 \text{ m}$ and $h = 3.6 \times 10^7 \text{ m}$; equatorial, rotating the same way as Earth.
- **Gravitational vs electric fields**: both follow $1/r^2$ (radial, infinite range); gravity only attracts and acts on mass.
- **Equipotentials** get further apart with distance because $V \propto -1/r$.
- **Astronauts DO have weight** ($g \approx 8.6 \text{ N kg}^{-1}$); they are in free fall with no contact force.

STARS AND SPACE

$\lambda_{\max}T = 2.898 \times 10^{-3} \text{ m K}$	hotter $\rightarrow$ shorter peak
$L = \sigma 4\pi r^2 T^4$	$r = \sqrt{(L/4\pi\sigma T^4)}$
$I = L/4\pi d^2$	I is measured, L is emitted
$z = \Delta\lambda/\lambda \approx v/c$	lab λ in the denominator
$v = H_0 d$; age $\approx 1/H_0$	age, not size; $\div 3.15 \times 10^7$ for years

- **Star chain:** Wien $\rightarrow T \rightarrow$ Stefan $\rightarrow L/r \rightarrow$ inverse square $\rightarrow I/d \rightarrow$ compare.
- **Solar rotation:** halve the edge-to-edge $\Delta\lambda$; $T = 2\pi r/v$.
- **H-R axes:** L/L_{Sun} (log) against T/K (**reversed log**, about 40 000 $\rightarrow$ 2500 K); Sun at 1 and about 6000 K.
- **Regions:** main sequence diagonal (mass increases upwards); red giants top right; white dwarfs bottom left (small, hot, dim, dense, no fusion); red dwarfs bottom right.
- **Old cluster:** red giants and white dwarfs present, top of the main sequence missing. Massive stars fuse H faster and leave first.
- **Standard candle** = object of **known luminosity** (not brightness; do not add "known distance").
- **Candle method:** locate it $\rightarrow$ **measure** $I \rightarrow$ known $L \rightarrow d = \sqrt{(L/4\pi I)}$. Dust lowers I , so d comes out too large.
- **Parallax:** 6 months apart (diameter of the orbit); angle against distant stars; know the orbit radius; **trigonometry, not Pythagoras**. Fails when the angle is too small.
- **Redshift** = fractional increase in received λ due to the source receding.
- **Absorption lines:** discrete levels; a photon is absorbed only if E exactly equals a gap.
- **Dark matter:** rotation speeds need more mass than is visible; it has mass but emits **no EM radiation** (not "no light").
- **Fate of the universe:** **density** vs critical density; uncertain because the amount of dark matter is unknown.

OSCILLATIONS

SHM DEFINITION (2 MARKS)

(Resultant) force/acceleration **proportional to displacement from the equilibrium position** and **always directed towards the equilibrium position**. Write "position", not just "equilibrium".

$a = -\omega^2 x$; $\omega = 2\pi f$	a - x straight line, gradient $-\omega^2$
$v_{\max} = \omega A$; $a_{\max} = \omega^2 A$	at equilibrium; at amplitude
$T = 2\pi\sqrt{(m/k)}$; $2\pi\sqrt{(l/g)}$	$k = mg/\Delta x$; use the total mass
$E_{k,\max} = \frac{1}{2}m(\omega A)^2$	or $\frac{1}{2}k\Delta x^2$

- **Phase:** v leads x by $\frac{1}{4}$ cycle; a and F are in antiphase with x . E_k - t has twice the frequency and is always ≥ 0 .
- **Resonance (3 marks):** driven at the natural frequency, so maximum energy transfer, so amplitude increases to a maximum.
- **Damping (2 marks):** work is done against resistive forces, so energy is dissipated to the surroundings and the amplitude decreases. It occurs at all frequencies.
- **Plastic/ductile materials** absorb energy and do not return it; elastic materials return it.

THE FOUR 6-MARKERS TO MEMORISE

A • GAS PRESSURE RISES ON HEATING

1. T up, so mean KE of molecules up
2. so speed and momentum up
3. Δp per collision with the **walls** up
4. rate of collisions with the walls up
5. rate of change of momentum up, so force up
6. $p = F/A$, so pressure up

B • CONDITIONS FOR FUSION

1. very high core temperature
2. so **nuclei/protons** have high KE
3. to overcome electrostatic repulsion
4. and get close enough to fuse (strong force)
5. very high density (gravity)
6. so the collision rate sustains fusion

C • EVOLUTION OF A SUN-LIKE STAR

1. main sequence: H fusion in the core
2. core H used up, so fusion slows/stops
3. core contracts under gravity
4. T rises **then** He fusion starts
5. outer layers expand and cool: red giant (top right)
6. fusion ends, core collapses: white dwarf (bottom left)

D • RESONANCE AND DAMPERS

1. people/wind drive the structure
2. driving f = natural f , so resonance
3. maximum energy transfer
4. amplitude rises to a maximum
5. energy is transferred to the dampers
6. dissipated (work against resistive forces), limiting amplitude

MCQ TRAPS

- The fate of the universe depends on density, not mass.
- A forced oscillator vibrates at the **driving** frequency.
- Amplitude $\times 2$ gives energy $\times 4$.
- B.E./nucleon rises in **both** fission and fusion.
- Absorption is exponential: 2 half-thicknesses give $\frac{1}{4}$.
- Same p , V , T : same N and mean KE; c_{rms} depends on mass.
- The H-R temperature axis is reversed; smaller λ_{max} means hotter.
- Higher I may just mean closer: I depends on L **and** d .
- Moving inwards, $\Delta E_{\text{grav}} = m\Delta V$ is negative.

CONSTANTS

G 6.67×10^{-11} · k 1.38×10^{-23} · σ 5.67×10^{-8} · Wien
 2.898×10^{-3} · c 3.00×10^8 · e 1.60×10^{-19} · u 1.66×10^{-27}
 kg · 1 MeV $1.60 \times 10^{-13} \text{ J}$ · 1 yr $3.15 \times 10^7 \text{ s}$ · $T(\text{K}) = \theta + 273$